

- 2 (a) State the *first law of thermodynamics*.

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- (b) A fixed mass of monatomic ideal gas undergoes a cycle of changes $A \rightarrow B \rightarrow C \rightarrow A$ as shown in Fig. 2.1.

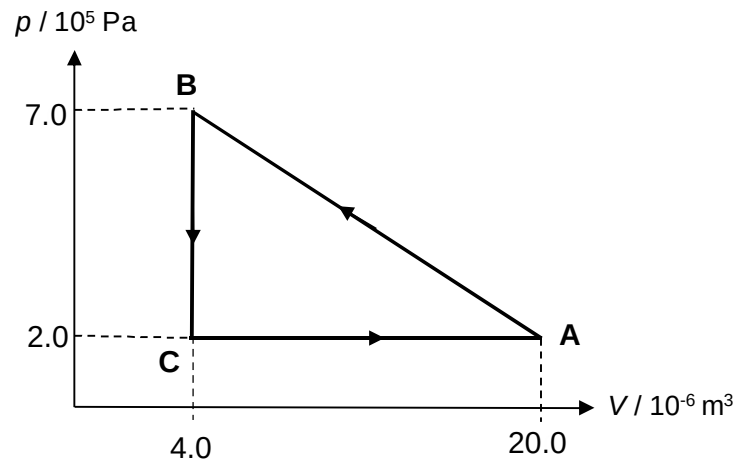


Fig. 2.1

- (i) Explain why the change in internal energy of the ideal gas during a complete cycle $A \rightarrow B \rightarrow C \rightarrow A$ is zero.

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- (ii) Calculate the work done on the gas during the process from A to B.

work done on gas = J [2]

- (iii) Calculate the change in internal energy of the gas when it undergoes the process from A to B.

Change in internal energy = J [2]

(iv) Calculate the heat supplied to the ideal gas during the process from A to B.

heat supplied = J [2]

(v) In a car engine, the expanding combustion gases push the piston which causes the car to move.

Suggest, with a reason, whether the cycle of changes $A \rightarrow B \rightarrow C \rightarrow A$ could be the cycle in the piston engine of a car.

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